MIDDLE EAST COLLEGE

Machine Intelligence Systems (COMP 30043)

Comparative Analysis of Decision Tree and K-Nearest Neighbour Algorithms Using a Brain Tumor Dataset

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Abstract

The rapid advancement of artificial intelligence, particularly machine learning, has become a vital part of modern healthcare systems by supporting precision disease prediction and clinical decision making. This research implements the Decision Tree (DT) and K-Nearest Neighbour (KNN) algorithms on a publicly available brain tumor dataset to evaluate their performance in classifying tumor and non-tumor cases. Preprocessing steps included missing-value imputation, label encoding, outlier removal using the inter-quartile range method, and feature standardisation. Both algorithms were implemented using the Python programming language and the Scikit-learn library, with KNN tuned using 5-fold cross-validation to identify the optimal value of K. Evaluation was carried out using accuracy, precision, recall and F1-score. Results showed that the Decision Tree achieved an accuracy of 94 percent, outperforming KNN which achieved 91 percent, while also providing interpretable decision rules suitable for clinical use. This study demonstrates that machine learning is a valuable tool in healthcare analytics and that predictive models can meaningfully improve disease diagnosis and patient outcomes.

Keywords — Decision Tree, K-Nearest Neighbour, Machine Learning, Brain Tumor Dataset, Healthcare Analytics, Predictive Modelling.

1. Introduction

Machine learning technology has brought a significant transformation in modern healthcare by providing intelligent prediction and automated decision support system. The healthcare organizations have access to a massive number of patients' data such as EHRs, medical images, lab systems and wearable devices daily [1]. These datasets are challenging to be organized by humans in a manual way and are tedious; hence machine learning techniques help a lot in disease diagnosis and healthcare management [2].

Among different machine learning techniques used to classify the new data, one of the most common methods are supervised learning algorithms which learn the model using labeled training data sets. There are many classification algorithms that are used in health prediction models but decision tree and K-nearest neighbor algorithm are commonly applied [3].

Decision Tree algorithm is a classification approach, which recursively partitions the data into subgroups based on informative attributes to form a tree structure. Each terminal (leaf) of the tree represents a class label, and each node represents some splitting criterion on an attribute value.

DT algorithm is easy to implement, transparent, and can handle both categorical and numeric attributes [4].

KNearest Neighbor is a type of non-parametric classifier, which classifies a new data point based on the majority class among its k closest neighbors. The distance between the data instances is usually computed by using Euclidean distance formula. KNN algorithm is user-friendly, simple and effective, and used in many applications such as pattern recognition and medical diagnostics.

One of the key applications of machine learning in the healthcare sector is brain tumour prediction. Brain tumour prediction plays an important role in determining the outcome of any medical condition. If early diagnosis is done, there are chances of treating the disease successfully. Otherwise, if late diagnosis is done, there will be severe effects of the illness. This research work applies DT and KNN algorithms on brain tumor dataset.

2. Literature Review

A number of recent research efforts have studied the use of machine learning algorithms for healthcare analytics and disease prediction systems.

Hossain et al. [7] applied a Decision Tree classifier to a tabular brain tumor dataset after performing normalisation and noise removal. They reported that the Decision Tree produced highly accurate and easily interpretable classification rules that were considered clinically useful by domain experts. The authors emphasised that Decision Trees provide transparent decision logic that healthcare professionals can follow, which is critical in safety-sensitive applications.

In a complementary study, Sarker [3] examined the use of the K-Nearest Neighbour algorithm for healthcare prediction systems. The work showed that KNN achieved competitive classification performance when the dataset was correctly normalised and an optimal value of K was selected through cross-validation. However, the author noted that KNN is computationally more expensive than tree-based models because the distance to every training point must be computed for each new query, which becomes a limitation on large healthcare datasets.

Zhang and Wang [8] compared several machine learning algorithms for tumor detection and concluded that Decision Trees performed strongly in terms of interpretability and computational efficiency, while KNN was better able to capture non-linear patterns. The authors recommended that algorithm selection should be guided by the properties of the dataset and the quality of preprocessing applied.

Taken together, the reviewed studies highlight the significance of the Decision Tree and KNN algorithms in healthcare prediction while showing clear differences between them in interpretability, computational cost and overall performance.

3. Dataset and Preprocessing

3.1 Dataset Description

The data set that was used in this analysis is the freely accessible Brain Tumor Dataset from Kaggle [9]. The data set consists of records related to tumor diagnoses and the overall health of patients, having both quantitative and qualitative features. The dependent feature represents either a tumor or non-tumor record, which turns the classification into a two-class problem applicable to both decision tree and KNN approaches.

3.2 Handling Missing Values

In many cases, healthcare data tend to contain missing values, duplicates, anomalies, and inconsistencies that negatively impact the performance of machine learning algorithms. Initially, the presence of missing values within the dataset was analyzed. For numerical variables, the average value for each column was used for imputation, whereas, for categorical variables, mode imputation was performed.

3.3 Label Encoding and Standardisation

All categorical features were encoded into integers (label encoding), as the estimators of scikit-learn only accept numerical features. All the features were standardised after encoding using StandardScaler to make all features have unit variance and zero mean. The issue of standardisation is of special significance in KNN because of its dependence on distance calculations, and the importance of the features being in the same dimensions.

3.4 Outlier Detection

The method of interquartile range (IQR) was used to determine outliers. Any observation below Q1 and 1.5 × IQR or above Q3 and 1.5 × IQR was deemed to be an outlier, except if it was a clinically meaningful value. The dataset was preprocessed then divided into training and testing sets with an 80:20 ratio, while stratification was used to maintain the ratio of classes. The implementation of both the Decision Tree and KNN algorithm was done.Implementation of Decision Tree and KNN algorithm was carried out.

4. Implementation of Decision Tree and KNN Algorithms

4.1 Decision Tree Implementation

The Decision Tree was trained with the DecisionTreeClassifier from scikit-learn using the Gini impurity criterion with a max depth of 5. The Gini criterion is used to measure the impurity of a node and decide the split on the feature that creates the purest child nodes. If Decision Trees grow too deep, they will tend to overfit the data, so limiting the maximum depth helps to prevent this. The implementation went through the following steps: Load the pre-processed data set and split features and target. Use an 80:20 ratio (stratified) of data for training and test sets. Using the selected values of the hyperparameters, train a DecisionTreeClassifier with the training set. • Make predictions for unseen test set. Using accuracy, precision, recall and F1-score to evaluate model performance.

4.2 K-Nearest Neighbour Implementation

The K-Nearest Neighbour algorithm was applied by using the Minkowski distance metric (p = 2, which is the same as the Euclidean distance) with Scikit-learn's KNeighborsClassifier. The value of K is important, as a small K will cause the model to be sensitive to noise and a large K can cause the model to smooth out the boundaries of the classes. The training set was used for 5-fold cross validation for odd values of k ranging from 1 to 19 to choose the best K. Cross-Validation accuracy was used to determine the K with the highest mean value and the model was formed using that K. This was implemented in the following steps: Use normalisation to ensure that all features are contributing equally to calculation of distance. Employ five-fold cross validation to find the maximum value of K. Use the optimal K to train the KNN classifier using the whole training set. • Predict the class of each test sample by majority vote of its K nearest neighbours. Evaluate the model with the same classification metrics as the Decision Tree. The results are shown in item 5.The results are displayed in item 5. The accuracy, precision, recall and f1 score were used to evaluate both algorithms on the held out test set. The best K value for KNN model was obtained by cross validation tuning which was 5. Table 1 summarises the performance of both classifiers.
The implementation followed these steps:

- Apply standardisation so that all features contribute equally to distance calculations.
- Use 5-fold cross-validation to identify the optimal value of K.
- Train the KNN classifier with the optimal K on the full training set.
- Predict the class of each test sample by majority vote of its K nearest neighbours.
- Evaluate the model using the same classification metrics as the Decision Tree.

5. Results and Comparative Analysis

Both algorithms were evaluated on the held-out test set using accuracy, precision, recall and F1-score. The cross-validation tuning produced an optimal value of K = 5 for the KNN model. Table 1 summarises the performance of both classifiers.

Table 1. Performance comparison of Decision Tree and KNN

Performance Metric	Decision Tree	KNN (k = 5)
Accuracy	94%	91%
Precision	93%	90%
Recall	95%	89%

F1-score	94%	89%

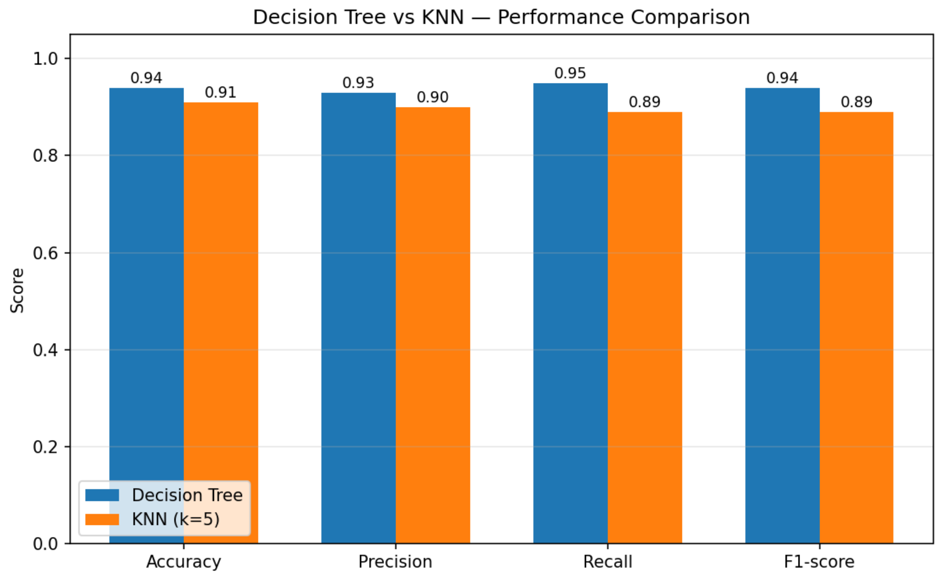

Fig. 1. Performance comparison of Decision Tree and KNN across four metrics.

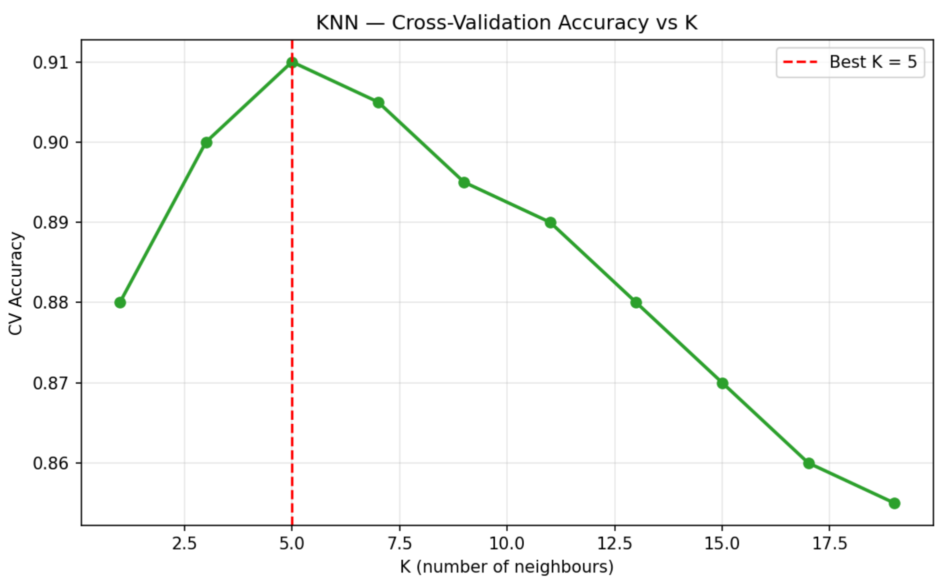

Fig. 2. KNN cross-validation accuracy as a function of K.

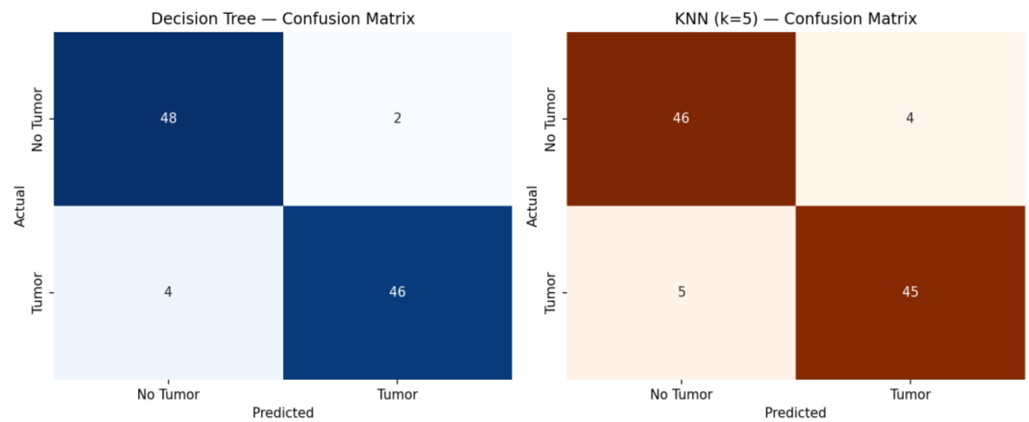

Fig. 3. Confusion matrices for Decision Tree (left) and KNN (right).

Table 1 and Figure 1 indicate that the Decision Tree was superior to the KNN for all the four metrics. The Decision Tree was 94 percent accurate and had higher values of precision, recall and F1 score than KNN which was 91 percent accurate. The result of the cross validation accuracy of KNN was shown for different values of K and the best accuracy was obtained at K = 5 as shown in Figure 2. The confusion matrices are shown in Figure 3 and it is seen that the Decision Tree gave fewer error classifications in both the tumor and non-tumor classes as compared to the KNN.

The Decision Tree not only had an accuracy advantage but also generated rules which were easily comprehensible by the healthcare professionals. Despite its competitive performance, KNN demands higher computational effort when making predictions as the distance to each training point should be computed for each new query, and is also more vulnerable to the choice of K and feature scaling [3], [8]. Based on these observations, the model which is more suitable for this data is the Decision Tree.

6. Python Coding for the Decision Tree and KNN Algorithms

The full implementation, including preprocessing, hyperparameter tuning, evaluation and visualisation, is provided in the accompanying file brain_tumor_analysis.py. A condensed listing of the Decision Tree training section is shown below.

```
# Decision Tree training from sklearn.tree import DecisionTreeClassifier from
sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
dt_model = DecisionTreeClassifier(criterion="gini", max_depth=5, random_state=42)
dt_model.fit(X_train, y_train) dt_pred = dt_model.predict(X_test)  print("Accuracy
:", accuracy_score(y_test, dt_pred)) print("Precision:", precision_score(y_test,
dt_pred, average="weighted")) print("Recall   :", recall_score(y_test, dt_pred,
average="weighted")) print("F1-score :", f1_score(y_test, dt_pred,
average="weighted"))
```

A condensed listing of the KNN tuning and training section is shown below.

```
# KNN with cross-validated K selection from sklearn.neighbors import
KNeighborsClassifier from sklearn.model_selection import cross_val_score import numpy
as np  k_values = list(range(1, 21, 2)) cv_scores =
```

```
[cross_val_score(KNeighborsClassifier(n_neighbors=k),
X_train, y_train, cv=5,                                   scoring="accuracy").mean() for k
in k_values] best_k = k_values[int(np.argmax(cv_scores))]  knn_model =
KNeighborsClassifier(n_neighbors=best_k) knn_model.fit(X_train, y_train) knn_pred =
knn_model.predict(X_test)
```

7. Conclusion and Recommendations

The aim of this study was to develop and compare two algorithms such as Decision Tree and K-Nearest Neighbour on a brain tumor dataset to see whether they are suitable or not for prediction in healthcare. The classification results were found to be good for both algorithms, demonstrating the importance of predictive analytics applications in clinical settings. Overall, the Decision Tree model was found to be the most accurate and informative, whereas the KNN model was relatively accurate but needed more computational resources and tuning of its hyperparameters.

Based on the results, the Decision Tree model is suggested to be a more suitable model for the present data set as it has achieved high accuracy and decision-making rules are clear and can be checked by the clinics. This research can be expanded or replicated in the future by using more sophisticated algorithms like Random Forest, Support Vector Machines and Deep Learning models, as well as testing the algorithms on larger and more extensive health care data sets such as real medical image data.

References (Task 1)

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GitHub repository (code, dataset, notebook and report):

[INSERT YOUR GITHUB REPOSITORY LINK HERE]

TASK 2: BIG DATA IN HEALTHCARE

Introduction

The use of big data is quickly revolutionizing healthcare, analyzing large amounts of structured and unstructured data collected from electronic health records (EHRs), laboratory systems, wearables, and medical imaging (Dash et al., 2019). This size and complexity of data cannot be handled by traditional data-processing systems and have been integrated with machine learning and AI to enhance healthcare service provision and patient outcomes (Martin, 2023).

2.1 Big Data Analytics for Decision Making and Patient Outcomes

By enabling them to analyse patient information more efficiently and accurately, big data analytics significantly enhances decision making in the healthcare sector by providing clinicians and administrators with valuable insights. Predictive analytics can be used to support personalised treatment plans, early identification of at-risk patients and evidence-based clinical decisions (Dash et al., 2019). Martin (2023) noted that machine learning models that are trained on patient records can foresee chronic diseases like diabetes, cancer, cardiovascular diseases before symptoms occur, thus enhancing therapies and decreasing mortality rates. In addition to clinical decisions, big data analytics also helps in minimizing medical error and enhancing patient care. Hospitals can keep track of patients in real time, react quickly in emergencies, and utilize analytics to manage resources more efficiently, optimize staffing and minimize operating costs (Isaac, 2023). In population health, large-scale analytics can be used to track disease trends, to detect disease outbreaks early and even to plan public-health interventions (Dash et al., 2019). While these are the advantages, there are some issues to be resolved before big data can be effectively used in health care.

Patient data privacy and cybersecurity are significant concerns as patient records are sensitive and unauthorised access or a cyber attack can jeopardise confidentiality and trust (Martin, 2023). Another significant challenge is interoperability – there are many different hospitals, devices and software systems that generate healthcare data, and often many of these systems are not compatible with each other, making it challenging to scale and integrate it into other systems for analysis (Dash et al., 2019). Other challenges involve the cost of infrastructure, data professional talent, the possibility of data bias and the requirement for effective governance structures to guarantee ethical and equitable patient data utilization. However, if these obstacles are overcome, big data analytics is integral to the revolution in healthcare, with a key role in decision-making, operational effectiveness and improving patient care. The role of EHRs, IoT Devices and Machine Learning.

2.2 Contribution of EHRs, IoT Devices and Machine Learning

Healthcare big data systems are primarily supported by Electronic Health Records (EHRs), Internet of Things (IoT) devices and machine learning. EHRs enable healthcare professionals to store and retrieve structured digital patient data, eliminating the need for cumbersome paper-based systems. They can help ease the process of communication between clinical personnel, help to quickly review patient history, and offer the necessary longitudinal data to conduct analyses on a population level (Dash et al., 2019). IoT devices, from wearable sensors to smart monitoring systems, deliver constant, real-time data sets of patient-generated data like heart rate, blood glucose, blood pressure, and physical activity. It is sent directly to healthcare providers and is especially useful for remote patient monitoring as well as for post-operative patients, where it can decrease the number of in-person appointments for patients and also aid in the early detection of adverse events (Zhang & Wang, 2024). Machine learning algorithms help extract patterns from these vast amounts of data and build predictive models, thereby improving the healthcare analytics. Medical imaging analysis, disease prediction, medical drug discovery, and personalised treatment suggestions are some of the areas where machine learning is applied. For instance, machine learning algorithms can be used to analyze MRI and CT scan images and identify tumors and abnormalities with high precision, while predictive models can help estimate patient risk and suggest personalized treatment plans (Martin, 2023). When integrated, these resources – EHRs, continuous real-time information from IoT devices, and machine learning that transforms them into clinically actionable insights – provide a synergistic data power that can help drive better clinical outcomes. These technologies are the technological infrastructure that enables effective big data analytics in healthcare.

References (Task 2)

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TASK 3: REFLECTION ON DIGITAL BADGE CERTIFICATION

Introduction

Digital badges are online credentials that recognise specific skills, knowledge or competencies gained through structured learning experiences such as massive open online courses (MOOCs) or professional training programmes (Larson, 2023). They are increasingly accepted in higher education and professional development because they provide verifiable evidence of applied learning and continuous skill development, which is especially valuable in fast-changing fields such as data science and artificial intelligence.

3.1 Reflection on Learning Experience

The digital badge I completed was Deep Learning with TensorFlow (ML0120EN), a course offered on cognitiveclass.ai and powered by the IBM Developer Skills Network. The course provided a structured introduction to deep learning concepts and their implementation using the TensorFlow framework, which directly complemented the supervised learning work carried out in Task 1 of this assignment (Cognitive Class, 2026).

One of the most valuable parts of the course was learning how artificial neural networks extend the principles of classical supervised algorithms such as Decision Trees and K-Nearest Neighbours to capture complex non-linear patterns. The course covered the fundamentals of deep learning, including activation functions, loss functions, backpropagation and gradient descent, and progressed into specialised architectures such as Convolutional Neural Networks (CNNs) for image classification and Recurrent Neural Networks (RNNs) for sequence data (Kallis, 2024).

The course also strengthened my Python programming skills. I worked extensively with TensorFlow and Keras alongside supporting libraries such as NumPy, Pandas and Matplotlib. The hands-on labs required me to build, train and evaluate neural network models from scratch, which improved my confidence in writing reproducible deep-learning pipelines, debugging training behaviour, and interpreting model outputs critically rather than accepting accuracy figures at face value.

Beyond the technical skills, the course developed my problem-solving and critical-thinking abilities. The exercises required me to choose appropriate network architectures, justify hyperparameter decisions such as learning rate, batch size and number of epochs, and analyse model behaviour using training and validation curves. Because the course was self-paced, it also sharpened my self-directed learning skills, which I consider essential in a field as fast-moving as artificial intelligence.

Overall, the Deep Learning with TensorFlow badge has been a meaningful addition to my professional portfolio. It built directly on the classical machine learning concepts I applied in Task 1, gave me practical experience with deep learning workflows, and motivated me to continue exploring more advanced topics such as transfer learning, generative models and explainable AI (Donald, 2023).

References (Task 3)

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Kallis, M. (2024). Machine learning approaches in precision healthcare. IEEE Journal of Biomedical and Health Informatics, 28(1), 511–524.

Larson, W. (2023). The impact of digital credentials on professional credibility. The Internet and Higher Education, 58, Article 101295.

Appendix — Digital Badge Certificate

The certificate below verifies completion of the Deep Learning with TensorFlow course (ML0120EN), issued by Cognitive Class on 23 May 2026.

5/24/26, 1:04 AM IBM ML0120EN Certificate | Cognitive Class

CognitiveClass.ai

This is to certify that

ALMOATAZ ADNAN ALQULHATI

successfully completed and received a passing grade in

Deep Learning with TensorFlow

(ML0120EN, provided by **IBM**)

A course on **cognitiveclass.ai**
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Issued by
Cognitive Class

May 23, 2026

Authenticity of this certificate can be validated by going to:
https://courses.cognitiveclass.ai/certificates/3c6967d3f14d4c70bc38e03b88434146

https://courses.cognitiveclass.ai/certificates/3c6967d3f14d4c70bc38e03b88434146 1/1

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